

# CHAPTER 5

## ROLE OF SPIKE ENERGY, HFD AND SHOCK PULSE (SPM) & SPECIFICATION OF THEIR ALARM LEVELS AT VARIOUS SPEEDS

### A. Spike Energy and Shock Pulse:

Ultrasonic instruments are designed to measure energy levels in roughly the 25,000 Hz to 100,000 Hz frequency range (1,500,000 - 6,000,000 CPM). These include Spike Energy<sup>1</sup> and Shock Pulse<sup>2</sup>. Each of these two parameters were developed in the 1970's and are designed to mechanically and electrically respond only to high frequency problem sources which excite the resonant frequency of their mounted transducer. Note the effective range for Spike Energy is from 5,000 Hz to 60,000 Hz, as shown in Illustration A. In the case of spike energy, the predominant accelerometer now in use (IRD 970) has a mounted natural frequency of about 30,000 Hz. Similarly, the reference mass within the shock pulse transducer is designed to respond at its resonant frequency of approximately 32,000 Hz.

Each of these technologies has proven worthwhile and capable of acting as effective tools at picking up initial stages of bearing wear if the user follows very strict measurement rules (these are covered in a Vibration Institute article entitled "The Use of Spike Energy for Fault Analysis and Machine Condition Monitoring" written by Joseph M. Shea and James K. Taylor of IRD). When bearings first begin to wear, they begin to excite natural frequencies of their components themselves (rolling elements, bearing races and cage) as discussed in the special rolling element bearing section in Chapter 6.07. One set of their natural frequencies is concentrated within the range of 500 to 2000 Hz (30,000 to 120,000 CPM). Another set is found within ultrasonic frequency ranges near the SPM and spike energy transducer natural frequencies. Therefore, when incipient wear just begins within rolling element bearings, the bearing components begin to impact exciting these natural frequencies which likewise excite the SPM and spike energy transducer natural frequencies. Basic defect mechanisms which generate ultrasonic response include:

- a. Micro spalls and cracks from fatigue or overstressing caused by brinelling, false brinelling, misalignment, overload, incorrect sealing and improper fits.
- b. Surface roughening from lack of sufficient lubrication.
- c. Surface indentations from hard contaminant micropitting from electric current passing through the bearing.

In the case of spike energy, accelerometers have been designed which have mounted natural frequencies on the order of 30,000 Hz (1,800,000 CPM) in the ultrasonic range. The sharp pulses and broadband random ultrasonic excitation from the bearings will excite the accelerometer natural frequency whether the accelerometer is stud-mounted, magnet-mounted, or probe-mounted (obviously, the stud-mount gives the highest spike energy magnitudes and most repeatable results). But all three mountings will provide spike energy measurements. This is somewhat surprising when one recalls what happens to vibration measurement quality results when he goes from stud to magnet to probe mount. However, like the empirical data that has been captured on rolling element bearings showing that their set of ultrasonic natural frequencies shows little change whether "free-free" or with the bearings mounted in the machine, the same

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appears to be true in the case of spike energy (Ref.1) . In "layman's" terms, spike energy is generated when the incoming ultrasonic frequencies excite the accelerometer's natural frequency, an electrical voltage response occurs which is then electronically conditioned and filtered, converting the end product to what might be called "impact energy" which is proportional to the incoming impact velocity. It is probably best thought of as a friction measuring parameter.

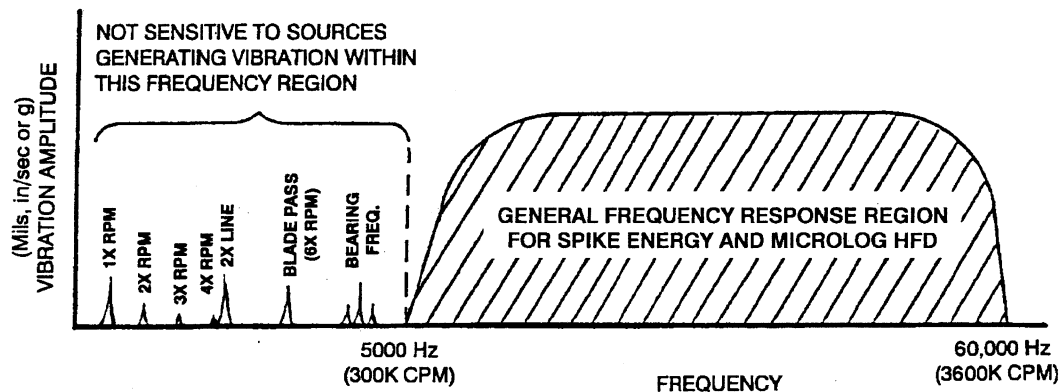
Spike energy pulses acting at high frequencies propagate rapidly through the structure. In the case of bearings, Reference 4 reports these elastic wave energy pulses travel through the bearing and surrounding structure at velocities approaching 16,000 ft/sec which is the velocity of sound (elastic) waves in steel (10,900 mi/hr!). When these impact waves encounter an interface of two materials or components, they lose much of their energy. This interface may be a region between the bearing outer race and its housing, or between one machine part and another to which this part is fastened. When these ultrasonic waves encounter an interface, some of the energy is transmitted through, and some is reflected back towards the source. The amount of energy reflected depends on a number of factors: the sharpness of the pulse, the difference in metals at the interface, how tightly the sources mate together, etc. In general, it is estimated that typically approximately 60% to 80% of the energy is reflected at each interface. **THEREFORE, THIS EXPLAINS WHY IT IS SO IMPORTANT TO KEEP THE MEASURING ULTRASONIC TRANSDUCER AS CLOSE AS POSSIBLE TO THE BEARING BEING EVALUATED.**

While this energy loss may be considered as a weakness, it also has an advantage. For example, when a bearing begins to fail, much of its vibration not only will exist within this bearing, but will transmit into other structures and bearings. On the other hand, ultrasonic energy is much more localized, making it easier to isolate the specific bearing having the problem.

Here, it is important to point out that not only does spike energy and shock pulse respond to bearing wear, but also such ultrasonic measurements respond to each of the following:

- bearing wear (as stated before)
- bearing lubrication
- cavitation
- rotor or seal rub
- belt squeal
- gear meshing
- sheave rub against a guard
- impact excitation of machine parts (i.e., valve in a reciprocating machine)
- steam and high pressure air flow

Since they do respond to all of these problems, this makes ultrasonic measurements of even greater use. In some cases, it might be confusing whether the problem is radiating from a bearing or from one of the other many sources listed above. However, if one remembers that ultrasonic energy dissipates rapidly (is very localized), it can help him in diagnosing which of several problems might exist. For example, in the case of a pump, if spike energy is high on both pump bearings, it is very likely that the pump might be experiencing cavitation or a seal rub. If the operator likewise records that he hears something like a "gravel sound", it might make a strong case for cavitation. On the other hand, if on the same pump, spike energy were high only on one of the two bearings, it would suggest possible wear or lubrication problems with this one bearing.



**ILLUSTRATION A.**  
**GENERAL FREQUENCY RESPONSE CURVE**  
**FOR SPIKE ENERGY AND MICROLOG HFD**

Finally, it is also important to point out what ultrasonic measurements do not respond to:

- unbalance
- misalignment
- bent shaft
- electrical problems
- eccentric rotors
- resonance
- structural looseness/weakness
- beat vibration problems

The reason that neither shock pulse nor spike energy respond to the above problems is that these problems tend to generate lower frequency vibration which is purposely filtered out in the design of the analyzer electronics [neither SPM nor spike energy are sensitive to problems which generate fault frequencies below 5000 Hz (300,000 CPM)]. Still, they are very useful in picking up the other problems such as bearing, lubrication, cavitation and rub problems.

Each of these technologies likewise have some disadvantages. In the case of shock pulse, the bearing bore and RPM (or bearing type and number) must be known in order to take its reference baseline (otherwise, the results are inaccurate). This is a disadvantage on many machines whose bearing model numbers and bores are not known. Also, since the shock pulse instrument usually makes only ultrasonic measurements, a second instrument is required to obtain vibration readings (particularly filtered vibration spectra) which will detect the lower frequency problems. In addition, since the shock pulse instrument is not yet included within any of the major predictive maintenance software offerings, its readings have to be manually entered into these PMP databases if one wants to both store and trend these measurements in his PMP software allowing him to directly compare vibration and shock pulse measurements in one trend graph. (However, SPM does now offer its own software for trending shock pulse readings themselves).

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In the case of spike energy, one of its disadvantages currently is that its amplitude response is highly dependent on which specific accelerometer (and mounting) is used to make the measurement. Amplitudes may differ by a factor of 5X to 10X from one accelerometer to another (however, if one always measures spike energy using the same accelerometer and mounting at identical locations from one survey to the next, meaningful trends and alarm levels can be established). One advantage spike energy has over SPM is that one does not have to know the bearing bore and model number to take a reading. However, another disadvantage has been found that when spike energy measurements are taken in the presence of steam, the steam itself can cause variant readings.

The advantages for both shock pulse and spike energy readings lie in the fact that they mandate one always take measurements at identical points from one survey to the next (not doing so will cause trends to vary widely). This will improve the accuracy, reliability and repeatability of the data from one survey to the next. In addition, if readings are taken with hand-held probes rather than magnet or stud mounted, simply a change in force on the transducer by the operator can raise or lower the subsequent reading. In both cases, if readings can be taken from stud-mounted or magnet mount locations, trends can greatly be improved. Spike energy measurements have also proven fairly repeatable using magnet mounts (particularly if the magnet exerts high force and can be mounted on a uniform surface), though not to the quality and repeatability of readings from stud mounts.

In any case, ultrasonic measurements are recommended on high-speed machines or lower speed machines having high frequency vibration sources (rolling element bearings, gears, etc). They can prove very effective in picking up many problems that might otherwise go unnoticed, particularly during incipient stages, but sometimes even in more advanced stages. In the case of bearings, each of these tools will pick up problems during the first of four definable stages of degradation (Ref. 9 & 10). Finally, meaningful severity charts are now being developed for each of these ultrasonic parameters. Figures 1 and 2 are included as example severity charts for spike energy and shock pulse, respectively.

### **High-Frequency Acceleration (HFD)**

Some vendors have offered high-frequency acceleration measurements in addition to vibration with their predictive maintenance hardware and software systems. While this is similar to the ultrasonic measurements mentioned above, this is truly a different measurement parameter altogether. In fact, as the name implies, this is a banded high frequency acceleration measurement which measures the total energy existing between certain lower and higher cutoff frequencies. Most often, this high-frequency measurement is referred to as HFD and is marketed by some of the major predictive maintenance vendors (see "Endnotes " 3 & 4). Figure 2 shows severity levels for "Microlog HFD" as compared with those of spike energy and shock pulse. In general, note that Microlog HFD levels tended to be approximately 2 times those of spike energy (that is, about 100% higher) in this particular study.

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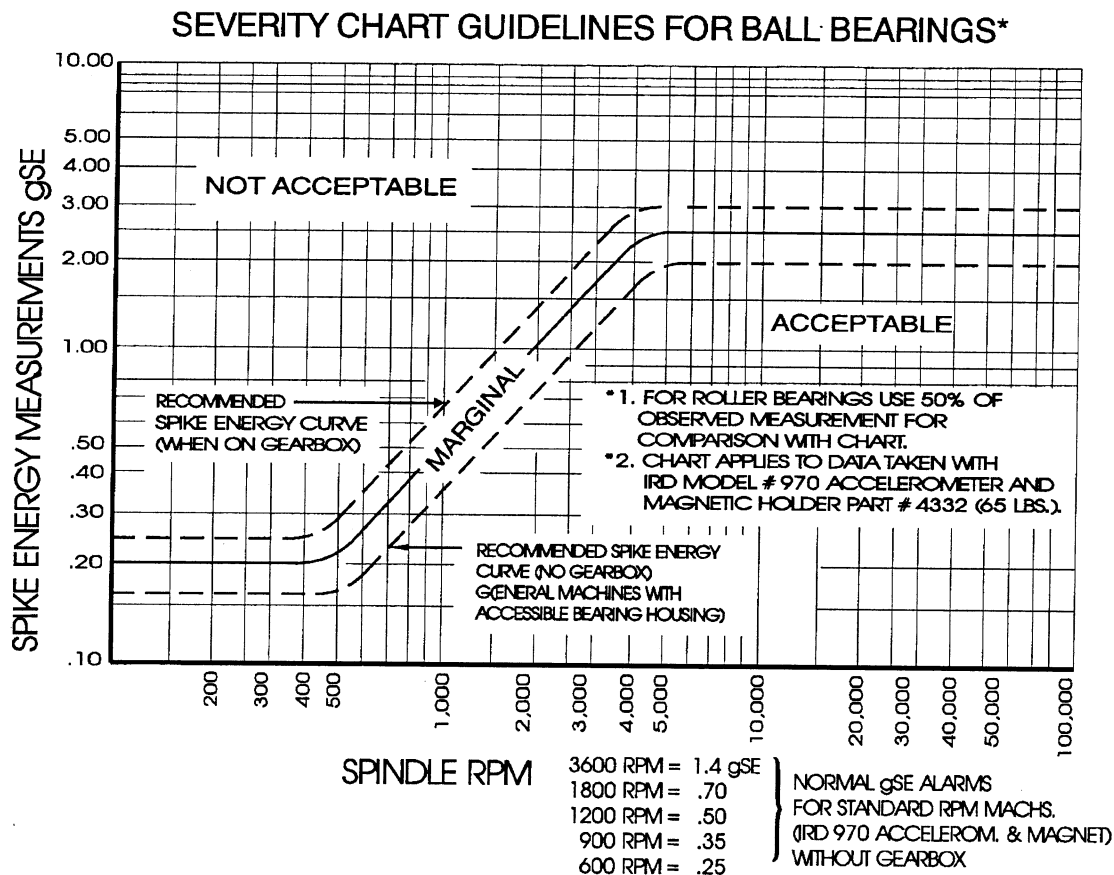
The thing which differentiates HFD measurements from those of shock pulse and spike energy is that HFD has not generally been an ultrasonic measurement, but instead a banded acceleration measurement. HFD by Vendor 3, is normally banded between approximately 5000 Hz and 20,000 Hz (300,000 - 1,200,000 CPM). However, this particular Vendor allows the user to specify lower and higher cutoff frequencies up to as high as 20,000 Hz. In these cases, it is understood that some users have specified their lower cutoff frequency as low as 1000 Hz up to 5000 Hz and their higher cutoff frequency anywhere from 3000 Hz to as high as 20,000 Hz. Microlog HFD, by Vendor 4, provides a fixed frequency range of 5000 Hz to 60,000 Hz (300,000 CPM - 3,600,000 CPM).

Like spike energy and shock pulse, HFD is sensitive to faults generating high frequencies such as bearing wear, gear wear, cavitation, etc. Due to its high frequency banding, HFD is not sensitive to problems such as unbalance, misalignment, eccentricity, etc.

#### **ENDNOTES**

1. "Spike Energy" was developed, patented and marketed by IRD Mechanalysis, Inc. based in Columbus, OH.
2. "Shock Pulse" was developed and marketed by SPM Instruments out of Marlborough, CT.
3. "HFD" was developed and marketed by CSI based in Knoxville, TN.
4. "Microlog HFD" was developed and marketed by SKF Condition Monitoring based in San Diego, CA.

**FIGURE 1**  
**RECOMMENDED SPIKE ENERGY SEVERITY CHART (IRD SPIKE ENERGY)**



### SPIKE ENERGY™ MEASUREMENTS

Energy is generated by repetitive transient mechanical impacts. Such impacts typically occur as a result of surface flaws in rolling-element bearings or gear teeth.

This energy is conducted from its source through various paths to the outer surface of the machine structure, and is seen as a small-amplitude vibration at the surface. Accelerometers coupled to the surface generate corresponding electrical signal.

The accelerometer signals processed by unique filtering and detection circuitry to produce a single "figure of merit" related to the intensity of the original impacts. This figure of merit is expressed in "gSE" units.

SPIKE ENERGY™ gSE readings are measurements which can with experience, be correlated with the severity of the casual surface flaws. Even though gSE readings are affected by the nature of the conductive path between the impact source and the accelerometer, similar machine structures will provide a reasonable basis for comparison between the structures.

The gSE figure of merit has proven to be effective in detecting mechanical defects in meshing gears and rolling element bearings. The gSE measurement, when used in conjunction with conventional measurement of vibration velocity and acceleration, provides early indications of mechanical deterioration.

**\*\*** When used with magnetic holders, accelerometers must be installed with a light coating of silicone grease and tightened to 40 in-lb. torque.

**FIGURE 2**  
**MAINTENANCE DIAGNOSTICS VIBRATION AND HIGH FREQUENCY**  
**GENERAL TOLERANCE CHART FOR PROCESS MACHINERY WITH**  
**ROLLING ELEMENT BEARINGS<sup>①</sup>**

| Unfiltered<br>Vibration<br>Velocity | Shock<br>Pulse &<br>Acoustic<br>Emission | Microlog<br>HFD | IRD<br>Spike<br>Energy <sup>②</sup> | Severity Level<br>and<br>Alarm Status |
|-------------------------------------|------------------------------------------|-----------------|-------------------------------------|---------------------------------------|
| Inches/Second<br>Peak               | Db                                       | g               | gSE                                 |                                       |
| 1.5+                                | 50+                                      | 5.0+            | 3.0+                                | Danger<br>(Shutdown)                  |
| .75 - 1.49                          | 40 - 49                                  | 3.0 - 4.99      | 1.50 - 2.99                         | Very Rough<br>(Alert)                 |
| .40 - .74                           | 30 - 39                                  | 1.50 - 2.99     | .80 - 1.49                          | Rough<br>(Alert)                      |
| .20 - .39                           | 20 - 29                                  | .75 - 1.49      | .40 - .79                           | Fair<br>(Acceptable)                  |
| .10 - .19                           | 10 - 19                                  | .30 - .74       | .20 - .39                           | Good<br>(Acceptable)                  |
| .01 - .09                           | 1 - 9                                    | .01 - .29       | .01 - .19                           | Smooth<br>(Acceptable)                |

① Provided by Charles Berggren of Monsanto

② Spike Energy Amplitudes measured using an IRD 970 accelerometer outfitted with IRD's 2-pole, 65 lb. magnet.

